

Right-Handed Neutrino Dark Matter

$$i\bar{\nu}_R \not{\partial} \nu_R - \bar{l}_L Y_\nu^\dagger \nu_R h^{(c)} - \frac{1}{2} M^\dagger \bar{\nu}_R^{(c)} \nu_R + h.c. \quad (1)$$

- Stabilize one of the right-handed neutrinos under a $\mathbb{Z}_2$ symmetry $e_R^{(1)} \rightarrow -e_R^{(1)}$.
- This sets the first row of the **PMNS matrix** Y_ν to 0, killing the weak interaction between $e_R^{(1)}$ and l_L .
- The right-handed neutrino dark-matter can still have a **Majorana mass**, but the corresponding left-handed neutrino is predicted to be massless.

Probing the Planck Scale: A Radiating Big-Bang

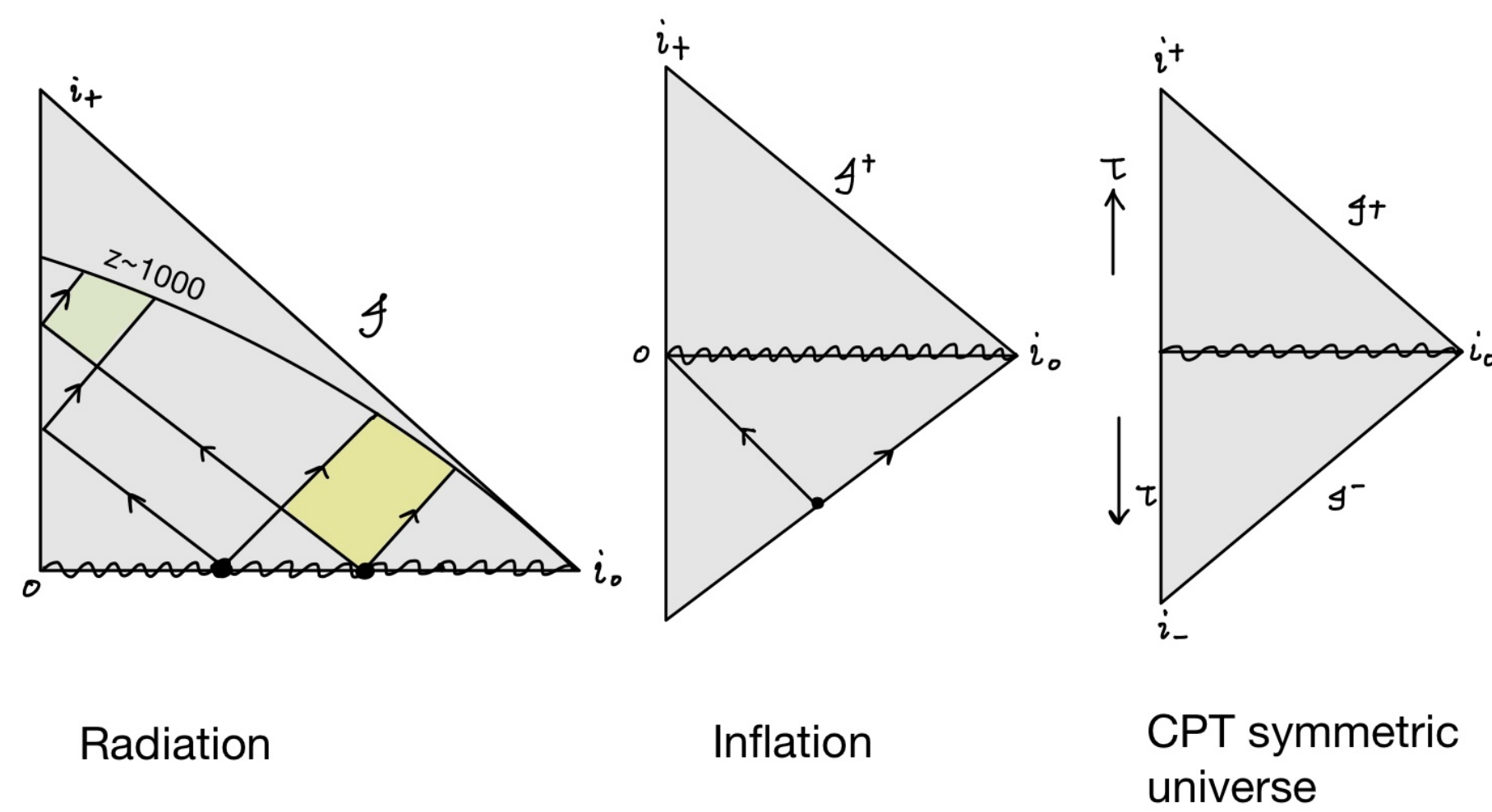


Figure 1. Penrose diagram for a Universe that is a) radiating, b) starts with inflation and then radiates, c) radiating with CPT symmetric boundary conditions.

Dark Matter Production in Quadratic Gravity

Quadratic Gravity Action:

$$S = \int d^4x \sqrt{-g} \left[M_{pl}^2 \frac{R}{2} + \frac{R^2}{6f_0^2} - \frac{1}{2f_2^2} C^2 + \Lambda \right] \quad (2)$$

- Renormalizable theory of gravity.
- Ghosts?
- Asymptotically Free or Landau Pole?

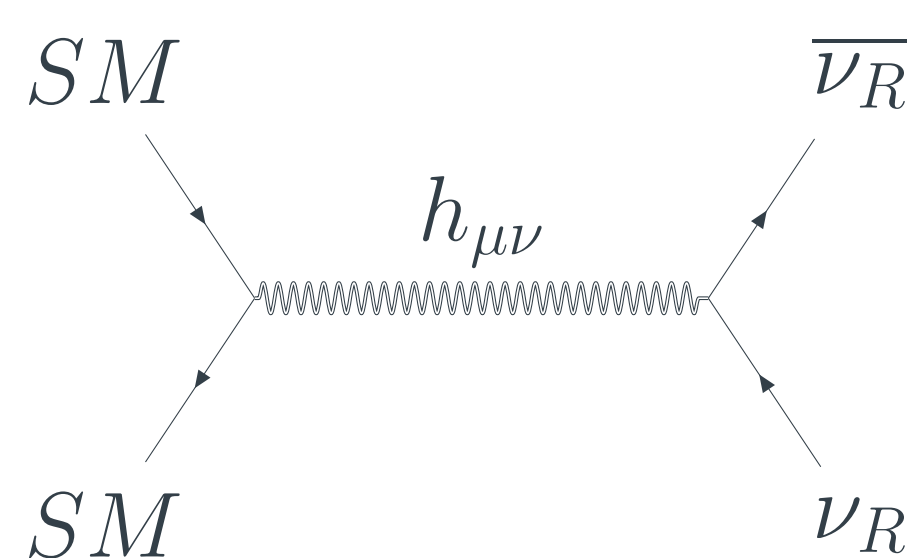


Figure 2. Standard Model particles gravitationally annihilating into right-handed neutrino dark matter. Such an interaction is suppressed in the infrared by the Planck Mass but becomes important when the temperature of the Universe approaches the Planck scale.

- The graviton propagator in this theory is given by:

$$\mathcal{D}_{\mu\nu\alpha\beta}(k) = \frac{4i}{M_{pl}^2} \left\{ \frac{\mathcal{P}_{\mu\nu\alpha\beta}^{(2)} - \frac{1}{2}\mathcal{P}_{\mu\nu\alpha\beta}^{(0)}}{k^2} - \frac{\mathcal{P}_{\mu\nu\alpha\beta}^{(2)}}{k^2 - M_2^2} + \frac{\frac{1}{2}\mathcal{P}_{\mu\nu\alpha\beta}^{(0)}}{k^2 - M_0^2} \right\} \quad (3)$$

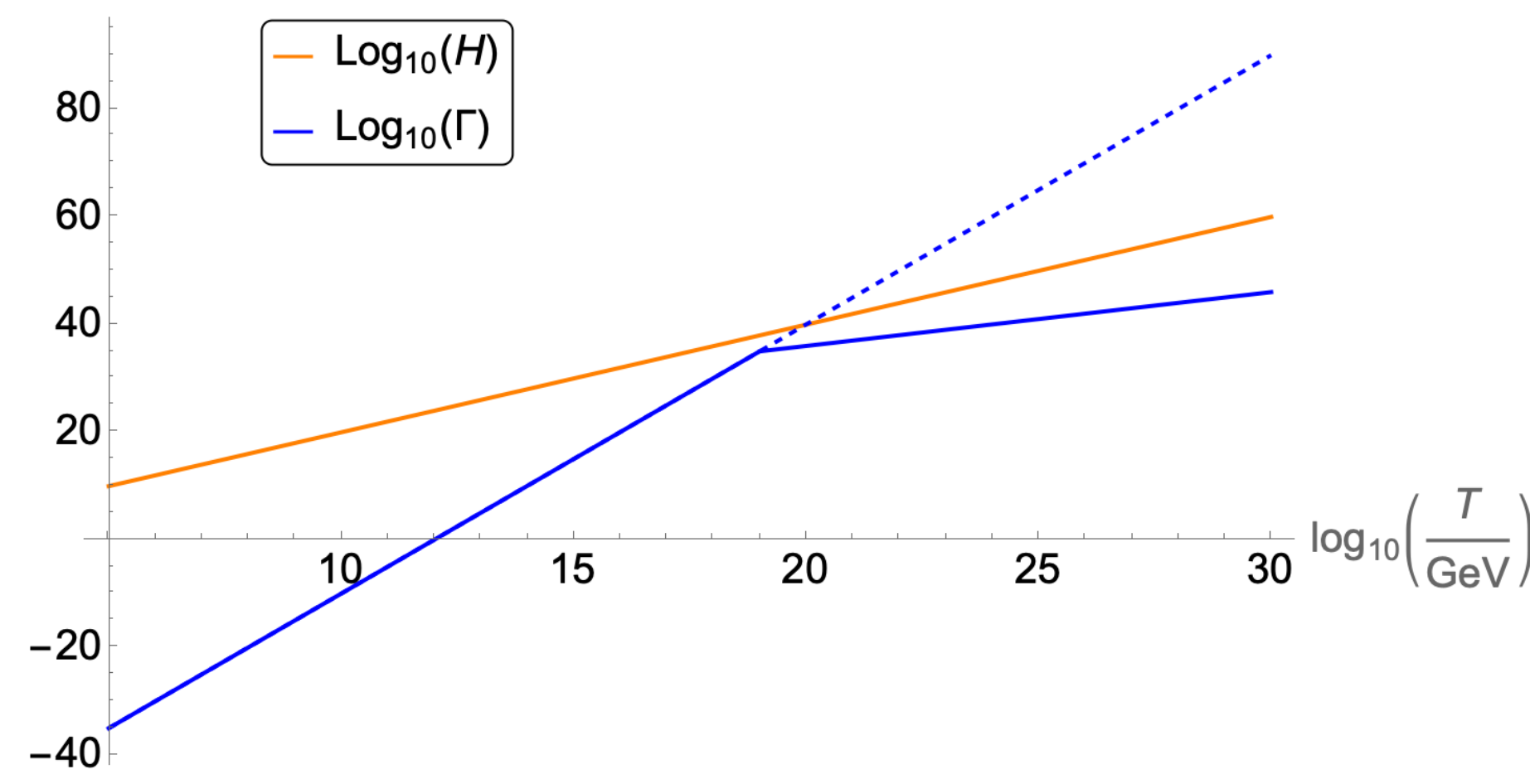


Figure 3. Comparison of the gravitational annihilation rate $\Gamma(T)$ gravity with the Hubble rate $H(T)$. In Einstein gravity (dashed blue line), $\Gamma(T) > H(T)$ (and unitarity is violated) when $T \gtrsim M_{pl}$. In quadratic gravity (solid blue line), the unitarity bound $f_2 \lesssim \mathcal{O}(1)$ implies $\Gamma(T) \lesssim H(T)$ for all T , and $f_2 \ll 1$ implies $\Gamma(T) \ll H(T)$, so that dark matter never thermalizes.

$$\Gamma(T) = \begin{cases} A_{IR} M_{pl}^{-4} T^5 & (T \ll f_2 M_{pl}) \\ A_{UV} f_2^4 T & (T \gg f_2 M_{pl}) \end{cases} \quad (4)$$

$$H = \left(\frac{\pi^2}{90} g_* \right)^{1/2} \frac{T^2}{M_{pl}} \quad (5)$$

Other Channels of Production: Cosmological Unruh Effect

- Dark matter overproduction and unitarity violation
- Too warm to be dark matter

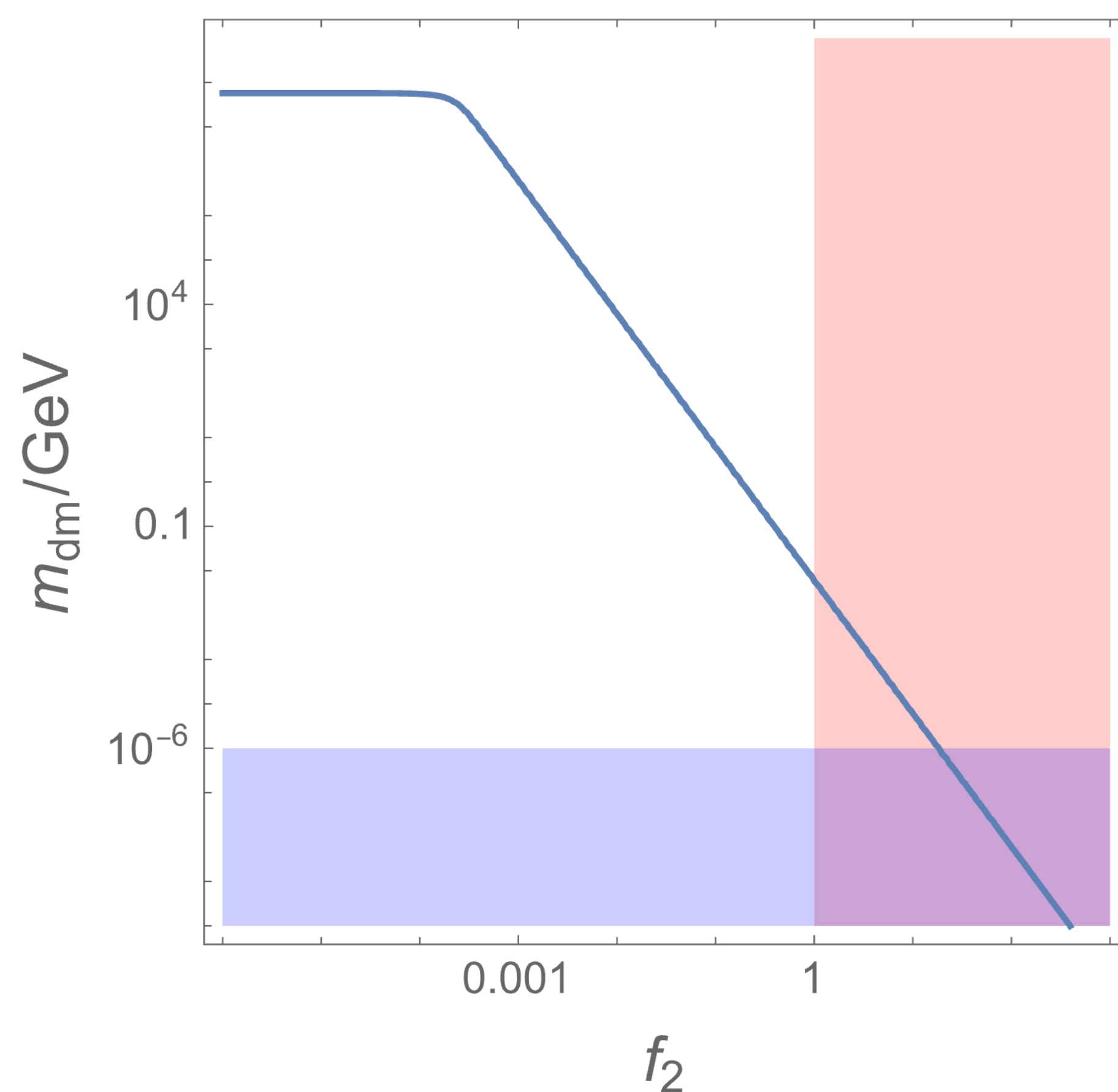


Figure 4. Constraint on the Weyl coupling f_2 and the dark matter mass m_{dm} coming from conservation of dark matter yield.

Taming the Biggest Explosion Ever!

- The Big Bang singularity occurs due to the scale factor. It would be smooth if the early Universe had **Weyl invariance**.
- Classically, the stress-energy tensor is traceless for a radiation dominated fluid. What about quantum effects?
- The **anomaly** is the violation of Weyl symmetry due to quantum effects. In four dimensions, it is captured by three coefficients:

$$\langle T_\mu^\mu \rangle = cC^2 - aE \quad (6)$$

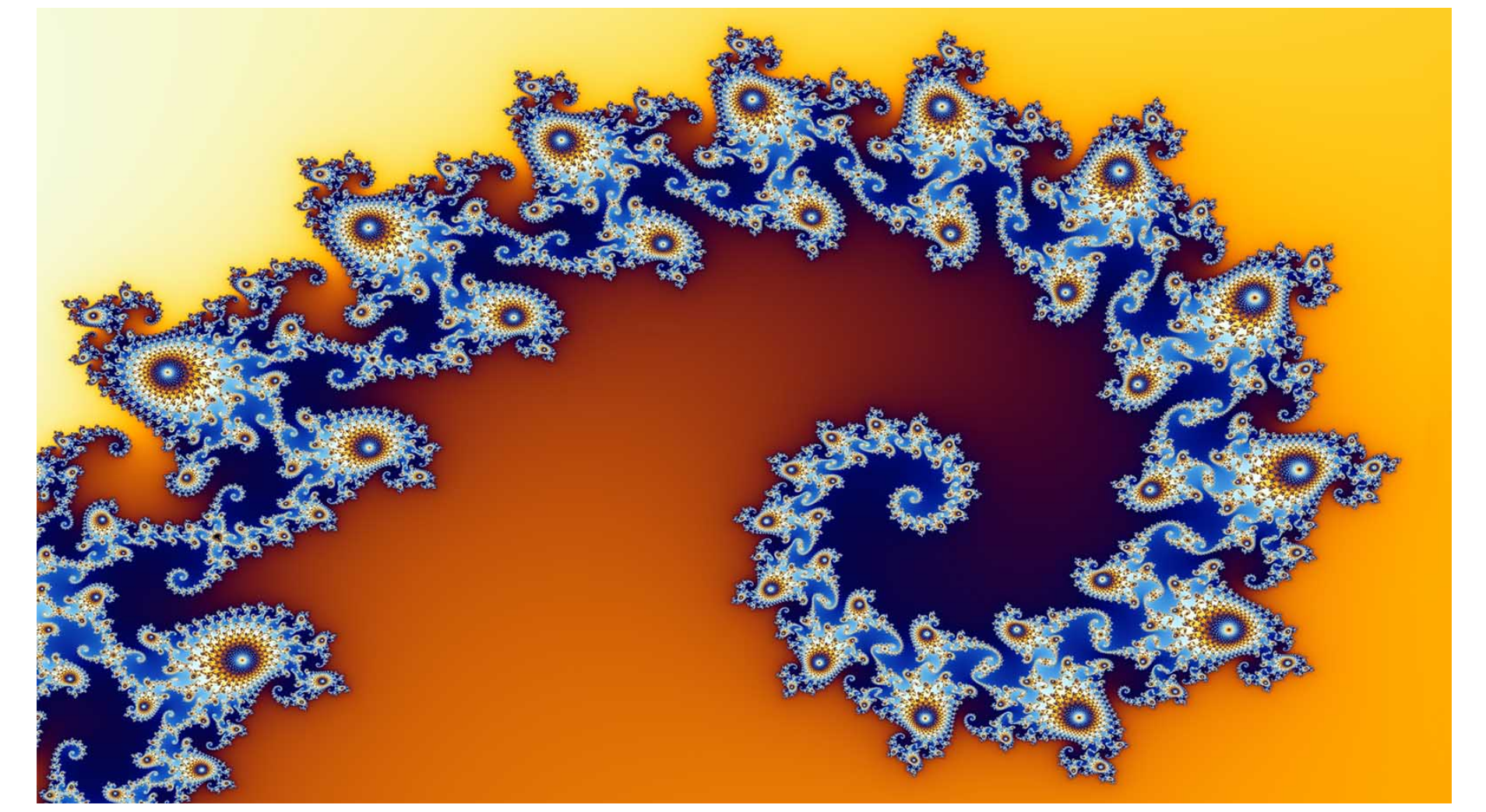


Figure 5. Weyl symmetry is the invariance of a theory under local rescaling. Look at this beautiful fractal, the design does not change, no matter how much you zoom in. [Source: Wikipedia]

- We use the **Exact Renormalization Group Equation** to calculate the a and c anomaly for the Standard Model fields:

$$\frac{d}{dt} \Gamma_k[\Phi] = \frac{1}{2} \text{Tr} \left(\left(\frac{\delta^2 \Gamma_k[\Phi]}{\delta \Phi \delta \Phi} + R_k \right)^{-1} \partial_t R_k \right) \quad (7)$$

- We find that in the presence of 36 Fradkin-Tseytlin scalar fields the a and the c anomaly vanish along with the vanishing of vacuum energy.
- This answers the **cosmological constant problem** by explaining why Λ is finite and not huge due to the quantum **vacuum energy**!
- The a and c coefficients are also the central charge for **stress-tensor correlators** that reek with UV divergences. Vanishing a and c coefficients means that the theory is well-defined in the UV and quantum gravitational correlators are finite.

References



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